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Goal: Prove that there are constants $C_1, C_2 > 0$ s.t.

$$C_1 \frac{x}{\ln x} < \pi(x) < C_2 \frac{x}{\ln x}, \quad x \geq 2.$$

Lower bound: Give $N \geq 2$, is there a naive choice of number that is divisible by every $p \leq N$?

Maybe $N!$, since every $p \leq N$ divides $N!$.

Moreover, $p > N \Rightarrow p \nmid N!$

Shortcomings. 1. $N!$ is too large!

2. May have $p^k \mid N!$ with k large.

Candidate: $\binom{2N}{N} = \frac{(2N)!}{N! \cdot N!} \in \mathbb{Z}$.

1. $p \mid \binom{2N}{N} \Rightarrow p \leq 2N$.

2. Size of $\binom{2N}{N}$ is more manageable.

3. There is a chance that large prime powers (excluding primes) do not divide $\binom{2N}{N}$.

Size

$$4^N = 2^{2N} = (1+1)^{2N} = \underbrace{\binom{2N}{0} + \binom{2N}{1} + \cdots + \binom{2N}{N} + \cdots + \binom{2N}{2N}}_{2N+1 \text{ terms}}$$

$$\Rightarrow \binom{2N}{N} < 4^N.$$

$\binom{2N}{N} \geq \binom{2N}{k}$, $0 < k < 2N$, $\binom{2N}{N} \geq \binom{2N}{0} + \binom{2N}{2N} = 2$

$$4^N = 2^{2N} = (1+1)^{2N} = \binom{2N}{0} + \binom{2N}{1} + \cdots + \binom{2N}{N} + \cdots + \binom{2N}{2N}$$

$2N-1$ terms $\leq \binom{2N}{N}$

$2 \leq \binom{2N}{N}$

So $4^N \leq 2N \binom{2N}{N}$, hence $\binom{2N}{N} \geq \frac{4^N}{2N}$.

$$\boxed{\frac{4^N}{2N}} \leq \binom{2N}{N} < 4^N, \quad N \geq 1.$$

$\geq 2^N$

Lem. If $N \geq 1$ and p is prime, then

$$e_p\left(\binom{2N}{N}\right) \leq \frac{\ln(2N)}{\ln p}.$$

Pf. Since $p > 2N$ does not divide $\binom{2N}{N}$, may assume $p \leq 2N$.

Start with

$$\binom{2N}{N} = \frac{(2N)!}{N! \cdot N!} = \frac{(N+1)(N+2)\cdots 2N}{N!}.$$

Need to show

$$e_p\left(\binom{2N}{N}\right) = e_p\left((N+1)\cdots(2N)\right) - e_p(N!)$$

$$\leq \frac{\ln(2N)}{\ln p}.$$

Choose $N+1 \leq m \leq 2N$ s.t. $e_p(m)$ is maximal.

Observe that $p^{e_p(k)} \mid k$ and $\underline{p^{e_p(k)} \mid p^{e_p(m)}} \mid m$

$e_p(k) \leq e_p(m)$

for all $N+1 \le k \le 2N$. By linearity,

$$p^{e_p(k)} \mid |m-k| , \quad N+1 \le k \le 2N.$$

Multiplying these, we get

$$p^{\sum_{\substack{k=N+1 \\ k \neq m}}^{2N} e_p(k)} \Bigg| \prod_{\substack{k=N+1 \\ k \neq m}}^{2N} |m-k| .$$

$$\prod_{k=N+1}^{m-1} |m-k| \cdot \prod_{k=m+1}^{2N} |m-k|$$

$$(m-N-1)! \qquad (2N-m)!$$

$$\binom{N-1}{m-N-1} = \frac{(N-1)!}{(m-N-1)!\,(2N-m)!} \in \mathbb{Z} \Rightarrow (m-N-1)!\,(2N-m)! \mid (N-1)!$$

$$\Rightarrow (m-N-1)!\,(2N-m)! \mid N!$$

So

$$p^{\sum_{\substack{k=N+1 \\ k\neq m}}^{2N} e_p(k)} \mid N!$$

$$\Rightarrow \quad \sum_{\substack{k=N+1 \\ k\neq m}}^{2N} e_p(k) \leq e_p(N!)$$

$$\Rightarrow \quad \sum_{k=N+1}^{2N} e_p(k) \leq e_p(N!) + e_p(m).$$

$$\sum_{k=N+1}^{2N} e_p(k) = e_p((N+1)\cdots 2N)$$

$$\Rightarrow \quad e_p((N+1)\cdots 2N) - e_p(N!) \leq e_p(m)$$

$$e_p((N+1)\cdots 2N) - e_p(N!) = e_p\left(\binom{2N}{N}\right)$$

Only need to show $e_p(m) \leq \frac{\ln(2N)}{\ln p}$.

$m \leq 2N$, $p^{e_p(m)} \leq m \implies p^{e_p(m)} \leq 2N$

$$\implies e_p(m) \ln p \leq \ln(2N)$$

$$\implies e_p(m) \leq \frac{\ln(2N)}{\ln p} .$$ □

Now,

$$\binom{2N}{N} = \prod_{p \leq 2N} p^{e_p\left(\binom{2N}{N}\right)}$$

$$\implies \underbrace{\ln \binom{2N}{N}}_{\geq 2^N} = \sum_{p \leq 2N} \underbrace{e_p\left(\binom{2N}{N}\right)}_{\leq \frac{\ln(2N)}{\ln p}} \ln p \leq \ln(2N) \underbrace{\sum_{p \leq 2N} 1}_{\pi(2N)}$$

$$\implies N \ln 2 \leq \ln(2N) \cdot \pi(2N)$$

$$\implies \pi(2N) \geq \ln 2 \cdot \frac{N}{\ln(2N)} = \frac{\ln 2}{2} \cdot \frac{2N}{\ln(2N)} .$$